

Power BI Basics – Quick Guide

Beginner Friendly ~20-Minute Read

Created for anyone who wants to grasp Power BI fundamentals fast — no fluff, just essentials.

Edition note: Based on Power BI Desktop interface as of 2026 (core views unchanged for years).

1 What is Power BI?

Power BI is Microsoft's powerful (and mostly free) tool that turns messy data (Excel sheets, databases, web data) into clear, interactive charts, dashboards and reports.

Why use Power BI?

- Stop staring at thousands of rows — see patterns instantly
- Spot trends, issues, opportunities in seconds
- Reports refresh automatically with new data
- Share professional-looking insights with team / management

Where companies actually use it (real examples):

- Sales → revenue by region/product, forecasts
- Finance → budget vs actual, margins, cash flow
- Marketing → campaign ROI, website traffic sources
- Operations → inventory, delivery performance
- HR → headcount, turnover rates
- → Almost every department that touches numbers!

2 Power BI Interface – The Three Main Views

Download **Power BI Desktop** (free) from <https://www.microsoft.com/power-bi> → open it.

Left sidebar has three icons — your main workspaces:

- **Report View** (home base)
Where you create charts, arrange pages, add filters/slicers. Drag fields from right pane → visuals appear.
- **Data View**
Table explorer — like Excel grid. Check raw values, sort, filter rows, see data types.
- **Model View**
Diagram of tables & relationships. Drag lines between columns (e.g. ProductID in Sales → ProductID in Products).

Quick tip: Click left icons to switch. Spend 80% time in Report View.

3 Getting Data In (Common Sources for Beginners)

Power BI connects to almost anything. Start simple:

- **Excel / CSV** → Most beginner-friendly
Home → Get Data → Excel / Text/CSV → pick file → Load or Transform.
- **SQL Database** → Get Data → SQL Server → enter server & credentials.
- **Web / APIs** → Get Data → Web → paste URL (JSON, HTML table, etc.).

Fast start: Search for “free sample sales data Excel” online → import it. After import → choose **Transform Data** (clean first) or **Load** directly.

4 Must-Know Basic Visuals (Charts)

In Report View → Visualizations pane (right) → pick these first:

- **Bar / Column Chart**

Best for *comparisons* (sales by city, top products).
→ Axis = Category, Values = Number field.

- **Line Chart**

Best for *trends over time* (sales by month, website visits).
→ Axis = Date, Values = Number.

- **Pie / Donut Chart**

Shows *proportions* (market share by category).
→ Use only with 5–6 slices. Legend = Category, Values = Number.

- **Card**

Big single number / KPI (e.g. Total Revenue = 15,67,890).
→ Drag one measure to it.

Quick format: Select visual → Format (paint roller icon) → change colors, title, labels, data labels.

5 The Core Power BI Workflow (Follow This Every Time!)

1. **Get / Import Data**

Home → Get Data → choose source → Load / Transform.

2. **Clean & Transform Data**

Fix errors (nulls, wrong types, extra columns) — do NOT skip!

3. **Model Data**

In Model view → create relationships (drag lines between tables).

4. **Build Visuals**

Report view → add charts → arrange layout → add slicers/filters.

5. **Analyze & Get Insights**

Look at visuals → ask questions → drill down → refine.

→ Save .pbix file often → Publish to Power BI Service (online) for sharing if needed.

6 Power Query – Quick Cleaning Intro

Click **Transform Data** after import → Power Query Editor opens (your cleaning friend).

Beginner actions:

- Remove blank rows → Home → Remove Rows → Remove Blank Rows
- Fix data types → Click column type icon → Whole Number / Date / Text
- Delete useless columns → Right-click → Remove
- Replace errors (e.g. "N/A" → 0) → Right-click → Replace Values
- Filter rows → Click column arrow → uncheck unwanted values

Rule: Always clean here — never change original file. When done → **Close & Apply**.

7 DAX Super Basics – Start With These

DAX = formula language for calculations. Go to Modeling tab → New Measure.

- **SUM** example:

Total Sales = SUM(Sales[Amount])

- **COUNT** example:

Number of Orders = COUNT(Sales[OrderID])

- **Simple IF** example:

Performance = IF([Total Sales] > 500000, "Good", "Needs Improvement")

Key: Measures auto-update with filters/slicers. Use them in visuals instead of raw columns.

8 Real Learning Tips (What Actually Works)

- Don't just watch videos — open Power BI and **do it immediately** after each step.
- Stuck? **Google it** — e.g. "Power BI remove duplicates Power Query"
- Interface may change slightly in updates — core stays the same.
- Start tiny: 1 file, 3 visuals, 1 page.

- Save versions often (File → Save As).

9 Top Beginner Mistakes (Avoid These!)

- Copy-pasting steps without understanding why
- Skipping data cleaning → wrong numbers everywhere
- Not understanding your source data columns
- Cramming too many visuals on one page — keep it clean & focused
- Overusing calculated columns — prefer measures for better performance

Mistakes are normal — fix them → learn → improve.

You're now ready!

Download Power BI Desktop → grab sample data → follow the workflow.
You've got this!

Screenshots Reference

Here are some visual examples to help you recognize the main parts of Power BI Desktop.

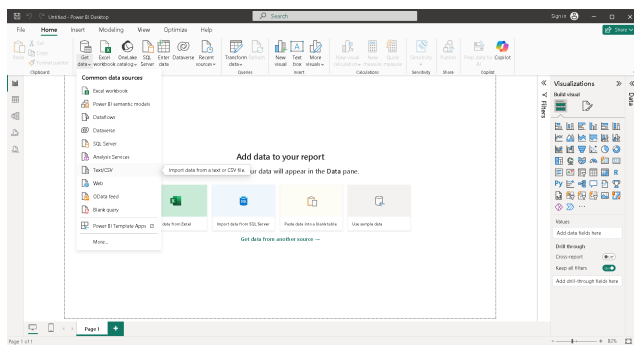


Figure 1: Power BI main interface

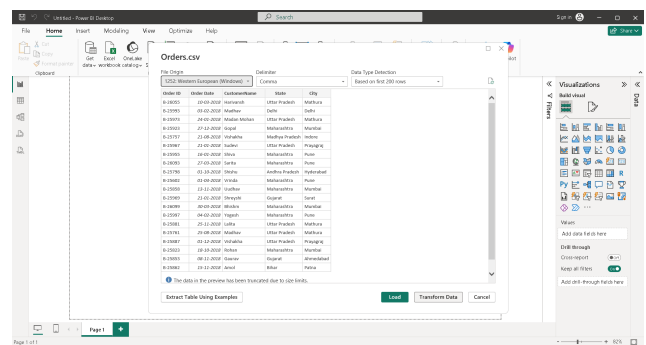


Figure 2: Importing Data in Power BI

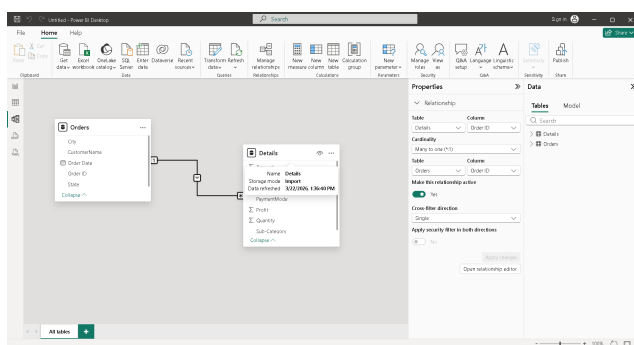


Figure 3: Model View – relationships diagram

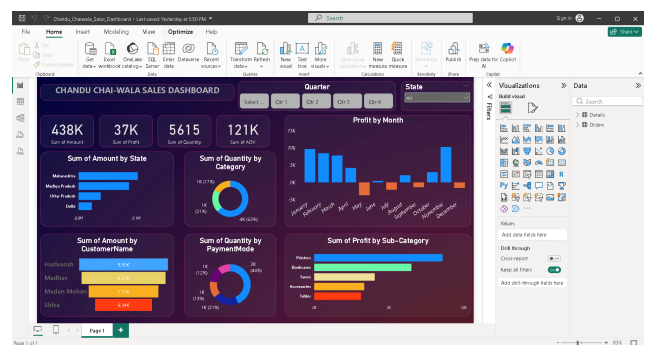


Figure 4: Example Dashboard For a small shop

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